

Taylor Expansion in 2 symmetric vars

Consider a function of 2 symmetric variables:

$$f(x, y)$$

Symmetry demands that:

$$f(x, y) = f(y, x) \quad f_x(x, y) = f_y(y, x) \quad \begin{aligned} f_{xx}(x, y) &= f_{yy}(y, x) \\ f_{xy}(x, y) &= f_{yx}(y, x) \end{aligned}$$

We are going to Taylor expand around

$$x = u_0, y = u_0$$

$$\begin{aligned} \text{Notation: } k_1 &= f_x(u_0, u_0) = f_y(u_0, u_0) \\ k_2 &= f_{xx}(u_0, u_0) = f_{yy}(u_0, u_0) \\ k_{11} &= f_{xy}(u_0, u_0) = f_{yx}(u_0, u_0) \end{aligned}$$

Let's do the expansion:

$$\begin{aligned} f(x, y) &= k_1(x - u_0) + k_1(y - u_0) + \frac{k_2}{2}(x - u_0)^2 + \frac{k_2}{2}(y - u_0)^2 + k_{11}(x - u_0)(y - u_0) \end{aligned}$$

Define the "symmetric" and "antisymmetric" vars $\delta u, v$:

$$\begin{aligned} \delta u &= (x + y)/2 - u_0 & v &= (x - y)/2 \\ &\text{or} & & \\ x &= \delta u + v + u_0 & y &= \delta u - v + u_0 \end{aligned}$$

When we do the substitution $f(x, y) \rightarrow f(\delta u, v)$:

$$f(\delta u, v) = 2k_1 \delta u + k_2 [(\delta u + v)^2 + (\delta u - v)^2] / 2 + k_{11} (\delta u + v)(\delta u - v)$$

$$= 2k_1 \delta u + k_2 (\delta u^2 + v^2) + k_{11} (\delta u^2 - v^2)$$

$$= 2k_1 \delta u + \delta u^2 \underbrace{(k_2 + k_{11})}_{K_+} + v^2 \underbrace{(k_2 - k_{11})}_{K_-}$$

$$= 2k_1 \delta u + K_+ \delta u^2 + K_- v^2$$

Now, we had insisted that at the Free energy minimum, $\delta u = (x + y)/2 - u_0 = (u_0 + u_0)/2 - u_0 = 0$.

Let's check.

$$\left(\frac{\partial f}{\partial \delta u} \right)_{\delta u=0} = 2k_1 = 0 \Rightarrow k_1 = 0$$

So, if $k_1 = 0$,

$$f(\delta u, v) = K_+ \delta u^2 + K_- v^2 \leftarrow \text{nice, right?? looks much cleaner than Eq [1]}$$

But it can be useful to write f in terms of $W = xy$ instead of $v = (x - y)/2$

In particular, it is going to be useful if you want an expression in terms of the arithmetic mean $\frac{x+y}{2}$ and the geometric mean $\sqrt{xy}$.

e.g. if the mean curvature $H = (k_1 + k_2)/2$ and the Gaussian curvature $K = (\sqrt{k_1 k_2})^2$.

If $w=xy$ and $\delta u=(x+y)/2-u_0$ and $v=(x-y)/2$

then an identity is:

$$w = (\delta u + u_0)^2 - v^2$$

We'll use this expression as a constraint, and then use the method of

"Lagrange multipliers"

$$\text{If } w = (\delta u + u_0)^2 - v^2 \text{ then} \\ [(\delta u + u_0)^2 - v^2] - w = 0$$

so, for any λ ,

$$\lambda [(\delta u + u_0)^2 - v^2 - w] = 0$$

"Lagrange multiplier"

add 0 to the expression for f :

$$f(\delta u, v) = 2k_1 \delta u + k_+ \delta u^2 + k_- v^2 + \lambda [(\delta u + u_0)^2 - v^2 - w]$$

and now we minimize:

$$\left(\frac{\partial f}{\partial \delta u} \right)_0 = 2k_1 + \lambda [2(\delta u + u_0)] = 0 \Rightarrow \lambda = -k_1/u_0$$

$$\left(\frac{\partial f}{\partial v} \right)_{v_0} = 2k_- v_0 - 2\lambda v_0 = 0 \Rightarrow \lambda = k_- \text{ or } v_0 = 0$$

however, we don't want to mandate that $v_0 = 0$, so we use the more general solution: $\lambda = k_- = -k_1/u_0$

So, rearranging $f(\delta u, v)$

$$f(\delta u, v) = \delta u^2 (k_+ + \lambda) + \delta u (2k_+ + 2u_0 \lambda) + v^2 (k_- - \lambda) - \lambda w + u_0^2 \lambda$$

but we know $\lambda = -k_-/u_0$ so $2k_+ + 2u_0 \lambda = 0$
and $\lambda = k_-$ so $k_- - \lambda = 0$

$$F(\delta u, v) = \delta u^2 \underbrace{(k_+ + k_-)}_{2k_2} - \underbrace{k_-}_{k_{11} - k_2} w - \underbrace{k_- u_0^2}_{\text{constant term, can be neglected}}$$

$$F(\delta u, v) = 2k_2 \delta u^2 + (k_{11} - k_2) w$$

or in terms of x and y

$$f(x, y) = \frac{k_2}{2} (x + y - 2u_0)^2 + (k_{11} - k_2)xy$$

$$\text{where } k_2 = \left(\frac{\partial^2 f}{\partial x^2} \right)_{u_0}, \quad k_{11} = \left(\frac{\partial^2 f}{\partial x \partial y} \right)_{u_0}$$

and u_0 is defined such that f is minimized
when $\frac{x+y}{2} = u_0$